

Quantum Gravitation

Problem Set 14

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Topic

Final Project: Joint Inference

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Final Project: Joint Inference. State all assumptions and conventions.

$$L = \chi_{scat}^2 + \chi_{prop}^2 + \chi_{therm}^2$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$accept = algebra \wedge registry \wedge resource$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$r_* = \operatorname{argmin}_r(L_r, d_r, \varepsilon_{num,r})$$

Show enough intermediate work that another student can reproduce the calculation.